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PwC builds multitenant enterprise decision support site with Power Pages

Executive summary

Big Four accounting firm PricewaterhouseCoopers (PwC) seized an opportunity to enhance service and build even more trust with its clients by migrating from its legacy client decision support system to an enterprise-grade, tech-enabled solution. The company chose to create a multitenant Microsoft Power Pages solution and was able to reimagine its data model by using Microsoft Dataverse with C# plugins to handle the business logic. In just six weeks, PwC successfully built the Cyber Technology Rationalizer site while gaining the benefits of low-code web development, including reducing costs and saving time. The team was able to reuse work done for the previous tool as code components in a Microsoft Power Apps component framework, which led to further time savings.

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Dushyanth Chandramouli, Technology Director, PwC

Solving more problems, faster

PricewaterhouseCoopers (PwC) has built its reputation on trust. As a Big Four firm and the second-largest professional services network in the world, PwC helps client organizations solve more problems while realizing more value—all with the goal of contributing to society. PwC considers helping clients solve important problems to be its way of making a lasting difference in the world. To that end, the company is determined to find new perspectives and new ways to solve old challenges.

One challenge that clients often bring to PwC is the question of decision-making, especially when it comes to cybersecurity. Clients of PwC rely on the organization's consultants and decision support systems to forecast projected costs for optimizing coverage across all pillars of the cyber security risk management and compliance framework. Previously, PwC helped clients with this process using a full-stack application based on React, Node.js, and PostgreSQL. But the tool was cumbersome and difficult to update, in part due to the way it incorporated business logic. PwC decided to build a scalable, enterprise-grade multitenant solution that not only addressed some of the weaknesses of the previous system, but would also provide customers with a new level of self-sufficiency.

An enterprise-grade solution

PwC had two priorities for the transition away from its legacy system. The first was to create a low-code solution. The original system included components that required specialized developers for time-intensive customization, so a low-code approach could improve maintainability, versatility, and scalability. The second priority was to enhance maintainability and change management: PwC intended to create one solution

that could work for many users, built with components that could be quickly reused and repurposed to speed innovation on future projects.

The Power Platform Center of Excellence team at PwC was up for the design challenge. The team wanted to build a multitenant solution that focused on business logic without all the initial setup and considerations a typical full stack project usually includes, like establishing a login or authentication.

Reimagining the data landscape

The first step was to build proof-of-concept solutions for the most impactful "unknowns." The team needed to discover how to integrate existing React components into the new system, to avoid duplicating all the work that went into building those components to begin with. "Once we proved the POC for a few components, that was a huge confidence booster," says Dushyanth Chandramouli, Solution Architect and Technology Director at PwC.

Another concern was the complexity of data manipulation and business logic computation in stored procedures in the legacy PostgreSQL database. After analyzing everything in the database, the team concluded that connecting to the PostgreSQL layer and reusing it was too complex. With a fresh perspective, the team decided to reimagine the data landscape. The team harnessed the Microsoft Power Pages Web API to interact with C# plugins in the Dataverse development layer for business logic computation.

With the unknowns answered, the Power Platform Center of Excellence team began building. In just six weeks, the Cyber Technology Rationalizer—the new PwC decision-making support solution—was complete.

Embracing the ecosystem

"As we've embraced the Microsoft ecosystem, a large part of that is embracing Microsoft Power Platform—the tools and capabilities that support our clients and engagement teams," says Karen Fraser, Program Operations Lead at PwC. Frayser says that Microsoft Power Platform helps PwC build high-value, impactful solutions in less time, at a fraction of the cost, with all the security, compliance, and data management benefits that come with Microsoft Azure.

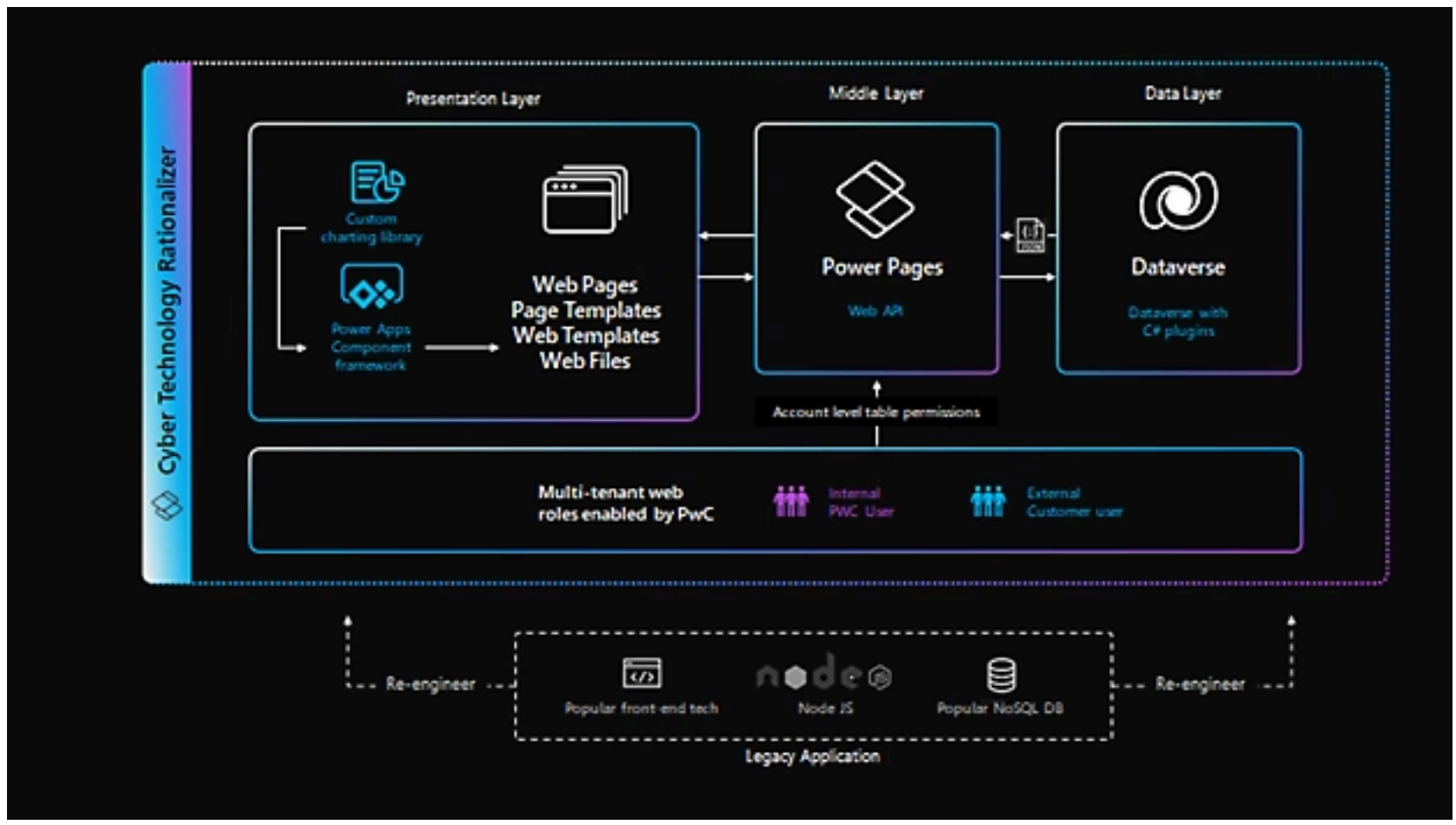

Figure 1. Diagram of the Cyber Technology Rationalizer architecture.

More about this diagram >

At the center of the PwC decision support solution is a website built with Power Pages. Users—both external customers and team members from PwC—log into the site. The multitenant features in Power Pages ensure secure access to only the appropriate data, as each user is associated with only one environment. While logged in, users provide information about their cyber licensing portfolio and can view snapshots of their existing coverage and costs. They can then model and analyze alternative scenarios at multiple levels, as well as take a deeper dive into potential opportunities to optimize costs and coverage.

The analysis and business logic computations are performed in C# plugins, producing JSON data which is then retrieved from Dataverse by the Power Pages Web API. The JSON data structure is transformed and integrated as Microsoft Power Apps framework components, ultimately rendering web-friendly visuals for the Cyber Technology Rationalizer.

Realizing more value

Since the launch of the Cyber Technology Rationalizer, PwC has achieved 85 percent cost savings and 30 percent time savings. "If you didn't know this was Power Pages, you might think it's a custom-developed web solution. It's working like that or better," says Chandramouli. "We've gotten very good feedback from the Cybersecurity, Risk, and Regulatory team, who are its primary internal users so far. Overall, it's been well-received and actively used in the external client marketplace."

The Cyber Technology Rationalizer helps CIOs and CTOs make strategic decisions to help improve cost and boost coverage, with a "What-if" analysis and other tools that can help users compare how certain decisions would likely affect the future. Because it allows users to manually upload new data and add additional rules, the site ultimately empowers users to explore questions on their own and make critical decisions with confidence. Solutions like the Cyber Technology Rationalizer demonstrate how PwC is converting full-stack developers to embrace and scale low-code as a part of the company's tech stack, transforming its operating model and driving its technology strategy.

Now that PwC has completed the Cyber Technology Rationalizer solution, the company is using its learnings to expedite future Power Platform-based solutions. Because the engineering team focused on reusability, elements of the solution like its user data management capabilities, master data structure, client management system, and data visualizations can all be repurposed and applied to future projects, helping decrease the time to innovation. With less dependence on developers and more building blocks to draw from, teams at PwC can focus on solving more problems for clients. "It's a trendsetter and baseline inspiration for what can come," says Chandramouli. "As someone who is very passionate about Microsoft Power Platform and the Microsoft ecosystem, the mission proved that we could build an enterprise solution—and by virtue of doing so, gain believers and build trust."

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